Deduce the World: How Far Can Reasoning Go?

Jeffrey Wei
Yale University

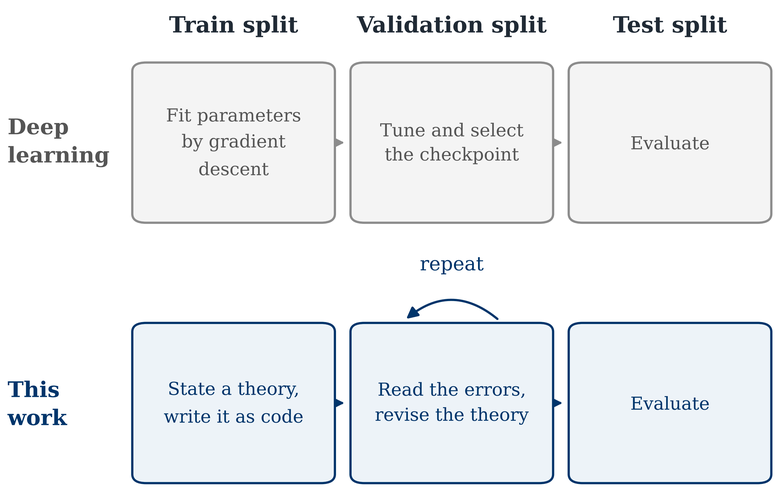
Brian Cheung
University of California, San Francisco

Contact
jeffrey.wei@yale.edu
jwei302.github.io

1 First principles versus deep learning

Science deduces the world from first principles, revising its theories where their predictions fail.

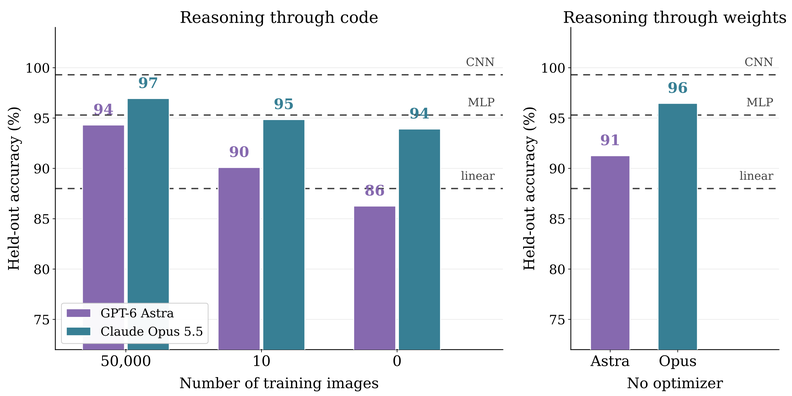
We ask whether a reasoning model can form scientific theories, and whether those theories generalize better than deep learning.



May	read papers and the web, measure the training data, inspect its own errors
May not	fit parameters to labels, search automatically over rules, store examples, see the test split

2 Classifying handwritten digits

Can a model classify handwritten digits from a stated theory of how digits are written, and how much data does it need to state one?

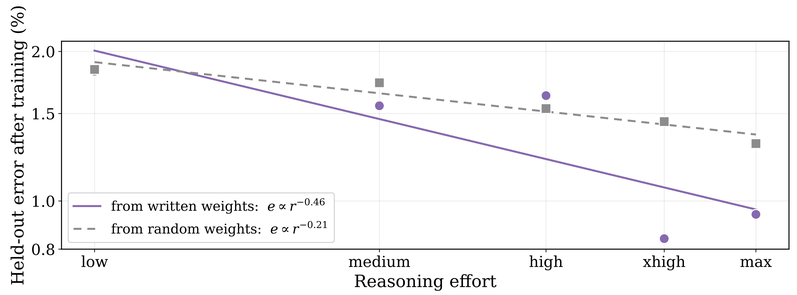


MNIST, 60,000 training and 10,000 test images. Dashed lines are trained models.

Given no training images at all, a theory of how digits are written classifies most of them.

Reasoning as initialization

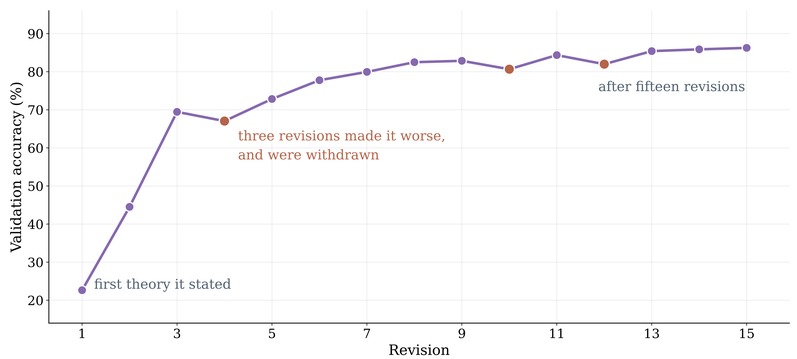
Can a model reason out an architecture and its weights directly, and is what it writes a better starting point than random initialization?



Networks initialized from written weights train to lower error than networks initialized at random, and the gap widens with reasoning effort.

The deduction–abduction loop

Can a model abduce why its own theory fails, and correct it?

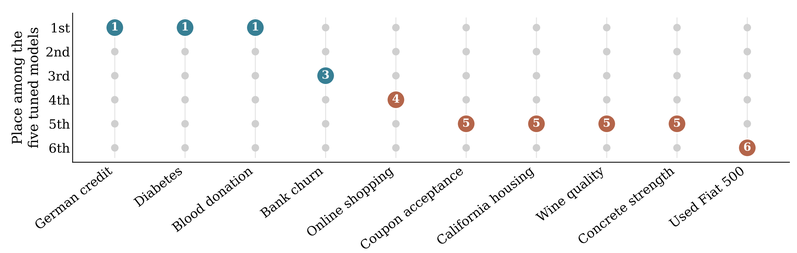


One digits session, every revision it made. Each began with a written hypothesis.

The deduction–abduction loop: abduce a cause, deduce its consequence, measure, and form the theory from what holds.

3 Reasoning on tabular data

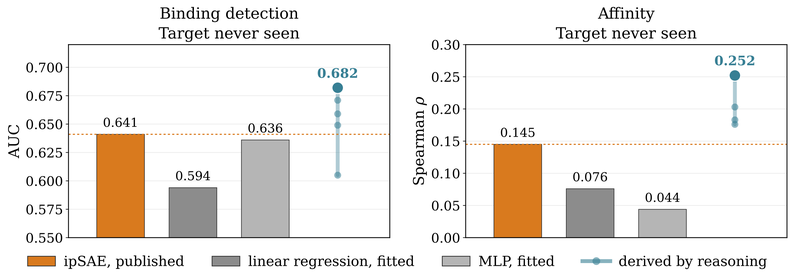
Does the same framework compete on tabular data?



Ten tasks from TabArena. Each ranks the reasoned theory against a linear model, a random forest, XGBoost, CatBoost and an MLP, all tuned on the same rows.

4 Protein binding evaluation metrics

Can a model derive a binding score from physics rather than fit one? The reasoning model receives a structure predictor's predicted aligned error (PAE) matrix.



704 designed binders, holding out a whole target. Detection on the left, affinity on the right.

On targets it never saw, the derived rule scores above the published score and above both fitted models.

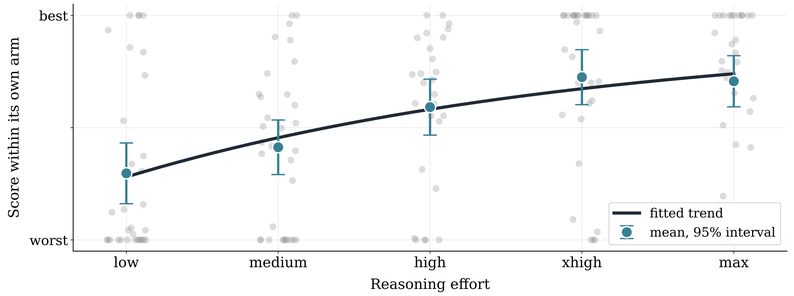
The stated theory

Treat the bound complex as a harmonic well. Equipartition makes the interface stiffness the inverse square of its fluctuation, and a stiffer well is a deeper one. Dividing by the binder's own internal error removes the predictor, and the units with it. Association confines six rigid-body degrees of freedom, each worth half the thermal energy times the log of the stiffness.

$$\kappa = \left(\frac{\sigma_0}{\sigma_{\text{iface}}}\right)^2, \qquad \mathrm{p}K_{\mathrm{d}} = \mathrm{p}K_{\mathrm{d,ref}} + 3\log_{10}\kappa.$$

5 Scaling reasoning effort

Does more reasoning produce better theories?



6 So how far can reasoning go?

We hypothesize that as language model reasoning improves, we will develop stronger first principles in the theory-poor, data-rich domains where deep learning currently dominates.